ALMONZER HAMID SARRAY ABDALLAH

Machine Learning Engineer | NLP & Large Language Model Researcher

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Languages: Arabic (Native), English (Fluent), Malay (Elementary)

RESEARCH PROFILE

Computer Science (Data Science) graduate of Albukhary International University on a full Albukhary Foundation scholarship, with a published Routledge book chapter, a medical-imaging preprint, and industry experience building and deploying machine learning systems under tight resource constraints. My work so far spans iris-based diabetes screening, offline OCR for handwritten Arabic prescriptions, Arabic dialect translation, LLM-based evaluation systems, and a source-grounded Arabic/English reading assistant. I am seeking a Master's by Research in Computer Science to study **Arabic and dialect-aware large language models**: how to evaluate them fairly across dialects and how to adapt them efficiently for low-resource, real-world use.

Research interests: Arabic NLP & dialects • LLM evaluation and alignment • parameter-efficient adaptation • efficient / on-device models • medical AI

EDUCATION

Bachelor of Computer Science (Honours), Specialisation in Data Science **Oct 2022 – Expected Oct 2026**
Albukhary International University (AIU) *Alor Setar, Kedah, Malaysia*

- **CGPA: 3.37 / 4.00** (111 credits earned). Full Albukhary Foundation Undergraduate Scholarship.
- **Relevant coursework (grade):** Artificial Intelligence (A), Probability & Statistics (A), Machine Learning (A−), Discrete Mathematics (A), Natural Language Processing (B), Deep Learning (B), Statistical Programming, Data Mining & Analytics, Data Structures & Algorithm Analysis.
- **Final Year Project (Project 1: A, Project 2: A):** MaiKedah, a smart-tourism analytics platform delivered to Majlis Bandaraya Alor Setar (see Projects).
- **Certifications:** ChatGPT Prompt Engineering for Developers (DeepLearning.AI & OpenAI) | Google Cloud Computing Foundations | Google Gemini Certified Educator.

PUBLICATIONS

- Z. Zainol, B. Gibba, A. A. B. Abdelgadir, **A. Abdallah**, M. S. Nabi, B. Riskhan (2026). **"Smartphone Use, Study Habits, and Academic Outcomes: Machine Learning Approaches to GPA Prediction."** In *Smart, Inclusive, and Sustainable Digital Societies: AI and Human-Centred Innovation* (Chapter 15). Routledge / Taylor & Francis. DOI: 10.1201/9781042012725-15
- M. M. Ahmed, **A. H. S. Abdallah** (2026). **"Detecting Diabetes Mellitus from Iris Imagery using Dual-Geometry CNN Ensembles."** Preprint, Zenodo. zenodo.org/records/19321167

RESEARCH & PROFESSIONAL EXPERIENCE

Afia Healthcare **Mar 2026 – Sep 2026**
Junior Machine Learning Engineer (Industrial Training) *Remote / Khartoum, Sudan*

- Built a dual-geometry EfficientNet-B3 pipeline for non-invasive blood-glucose screening from iris images (raw images and polar maps), reaching **92.86% accuracy** and **0.9464 ROC-AUC** on 56 held-out patients.
- Prevented data leakage across 196 clinical subjects with strict patient-level train/test partitioning, and anonymised medical images with SHA-256 hashing.
- Co-authored the resulting preprint, including the validation of non-destructive augmentation for iris-based diagnosis.
- Developed an offline OCR model for handwritten Arabic prescriptions (CRNN: ResNet + BiGRU + CTC loss) reaching **97.9% accuracy**, quantised to **15 MB** with INT8 TFLite/ONNX to run on low-end phone CPUs.

Appen (Malaysia) **May 2026 – Aug 2026**
AI Trainer *Remote, Malaysia*

- Trained 40+ multilingual annotators in data-quality practice, statistical sampling, and LLM safety-evaluation protocols.
- Wrote evaluation SOPs for LLM prompt alignment and risk queries, speeding up the annotation pipeline by about 25%.

SELECTED PROJECTS

Lenvra Read: AI Guided-Reading Web App **2026 – Present**
Founder & Product Builder *lenvra-read-app.vercel.app (live pilot)*

- Web app that turns a nonfiction book PDF into guided reading sections that follow the author's argument in order, with every point linked to the page it came from; full English and Arabic (right-to-left) support.
- Designed a three-step LLM pipeline (write, fact-check, repair) with structured JSON output and automatic checks that every cited page reference exists.
- Hybrid OCR for scanned books (cloud vision model plus on-device Tesseract) at about US$0.05 per 120-page book; chose models through a 14-section evaluation of accuracy, cost and speed.

Mihna AI: Voice-First Candidate Evaluation Engine **2025 – Present**
Lead AI Engineer *Production platform*

- Built an LLM evaluation pipeline that transcribes spoken interview answers (Whisper) and scores them against 10 competencies with an LLM, cutting review time by 68%.
- Added multi-model fallback routing (Llama 3.1 → Qwen → Llama 3.3) so screening keeps running through single-provider API outages.

SaySimple: Arabic Dialect Simplification & Translation **2025**
Google AI Hackathon (APAC) Finalist

- LLM web app that simplifies dense text and translates it into 50+ languages, including 22 regional Arabic dialects.

MaiKedah: Smart Tourism Analytics Platform **Aug 2025 – Feb 2026**
Lead Software Engineer, Final Year Project *Alor Setar, Malaysia*

- Flutter tourist app and TourStat dashboard for city administrators on a shared Supabase (PostgreSQL) backend; requirements gathered with Majlis Bandaraya Alor Setar and demonstrated at a regional government technology summit.

Other builds **2025 – 2026**

- **UltraFolio** (co-founder, 2025 – present): Next.js SaaS that turns a CV into an interactive 3D portfolio; 100+ active users. ultrafilo.vercel.app
- **Your Medicine Guide** (Code for Sudan Hackathon 2025): medicine and doctor locator for crisis zones in Sudan.
- **CodeTrace AI Bug Detective** (hackathon, 2026): AI tool that explains the root cause of code bugs; built in a team of four.
- **ScholarAI Navigator** (Bolt Hackathon 2025): matches a student's CV to scholarship opportunities for African youth.

TALKS, TEACHING & SERVICE

Invited Speaker, Weekly Diplomatic Majlis **Jul 2026**
Hosted by the ambassadors of Oman, Iraq, Qatar and Libya *Kuala Lumpur, Malaysia*

- Spoke on the foundations of modern AI (the Transformer and "Attention Is All You Need"), what today's models can do, and why the Arab world needs its own Arabic-language AI capacity.

Head of Academic Affairs **2023 – 2024**
Sudanese Students Association, AIU *Alor Setar, Malaysia*

- Ran technical bootcamps on machine learning, NLP pipelines and open-source tools for 50+ students; set up peer-mentoring groups.

Technical Workshop Facilitator **Dec 2025**
Malaysia Madani STEM Programme *Kedah, Malaysia*

- Taught 3D modelling to 70+ secondary-school students (ages 12–17) from low-income rural communities across two sessions.

AWARDS

- **National CTF Cybersecurity Tournament (2024):** 1st place, Round 1; 4th place, Finals.
- **Google AI Hackathon, APAC (2025):** Finalist (SaySimple).
- **Albukhary Foundation Undergraduate Scholarship (2022 – 2026):** full scholarship.

TECHNICAL SKILLS

- **ML & NLP:** PyTorch, CNNs (EfficientNet), CRNN/CTC, Whisper, TF-IDF, SVM, Random Forest; LLM APIs (Claude, Gemini, Llama, Qwen), Tesseract OCR, prompt engineering, LLM evaluation.
- **Efficient deployment:** INT8 quantisation, ONNX, TFLite, offline / on-device inference.
- **Statistics:** hypothesis testing, inferential statistics, ROC-AUC analysis, patient-level validation design.
- **Software:** Python, TypeScript, Next.js, React, Flutter, FastAPI, PostgreSQL, Supabase, Docker, Git, AWS SageMaker.